THEODORE HESLING

Engineering student seeking a 4-to-5-month internship in Embedded Systems, Robotics & Smart Sensors

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PROFILE

Systems Engineering candidate at Mines Saint-Étienne (ISMIN) specializing in embedded architectures, autonomous robotics, and sensor physics. Proven track record in prototyping complex mechatronic systems, bare-metal C programming on ARM Cortex-M, and designing custom capacitive sensors. Motivated to contribute to Swiss robotics, medical instrumentation, and precision industrial automation. EU citizen eligible for simplified Swiss L/B work permits.

EDUCATION

Engineer Degree (M.Sc) - Mines de Saint Etienne 09/2025 – 07/2028

Engineering Degree candidate at Mines Saint-Étienne (ISMIN cycle). Specialization in Embedded Systems and Autonomous Mechatronics. Core coursework: Microcontroller Architectures (STM32), Real-Time Operating Systems (FreeRTOS), Sensor Acquisition & Signal Processing, Feedback Control (PID), and 3D CAD. Gardanne

CPGE PC - Lycée Jeanne d'Albret 09/2023 – 07/2025

Intensive two-year undergraduate program in Advanced Physics, Chemistry, and Mathematics preparing for competitive nationwide entrance exams to French Engineering Grandes Écoles. SAINT GERMAIN EN LAYE

EXPERIENCE

Autonomous Boat Prototype (Self-Levelling) - Mines Saint-Étienne 10/2025 – 03/2026

Engineered a self-levelling catamaran using STM32 bare-metal C. Developed custom differential capacitive liquid level sensors measuring water-column dielectric variation ($\Delta\epsilon_r$), processed via ADC into a real-time PID stabilization loop driving ballast servomotors. Modelled mechanical chassis in Autodesk Inventor. Gardanne

Treasurer - Junior Mines Provence 01/2026 – 07/2026

Managing financial operations and budget allocations for engineering projects, tracking procurement for technical prototyping, and ensuring strict compliance with CNJE audit guidelines. Gardanne

Cybersecurity Platform Architect Intern - Valeo 01/2026 – 02/2026

Analyzed embedded bus communications (CAN/LIN) and evaluated real-time sensor data integrity in safety-critical electronic control units. Créteil

Observation Internship - BNP Paribas CIB 04/2021 – 04/2021

Observed mission-critical automated infrastructure, fault-tolerant system monitoring, and low-latency data telemetry in a global financial institution. Shanghai

SKILLS

Embedded Software
Skillful (C, Bare-metal STM32, ARM Cortex-M, FreeRTOS, HAL)

Control Systems & CAD
Skillful (PID Closed-Loop, Autodesk Inventor, Prototyping)

Python & Data Processing
Experienced (Python, NumPy, Matplotlib, Data Acquisition)

Sensor Design & Physics
Skillful (Capacitive Sensing, Dielectric $\Delta\epsilon_r$, Signal Conditioning)

Hardware Interfaces
Experienced (ADC, PWM, Timers, SPI, I2C, UART, CAN Bus)

Lab Equipment & Tooling
Intermediate (Oscilloscopes, Logic Analyzers, Multimeters, Soldering, Git)

LANGUAGES

French
Native speaker

Spanish
C1 – Fluent (DELE Diploma)

German
Elementary (A1/A2, self-study with Nicos Weg)

English
C2 – Bilingual (IGCSE & A-Levels)

Chinese (Mandarin)
C1 – Fluent (HSK 3)

Swiss Work Authorization
French / EU Citizen (Fast-track L/B Permit)